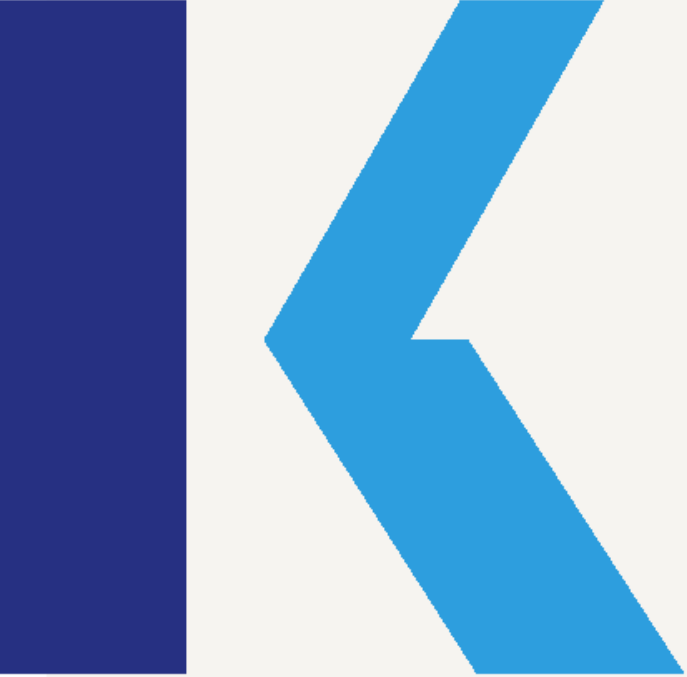




DATA4600 Business Analytics Project Management

Workshop 5

Analytics projects – managing time with
GANTT charts



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Learning Outcomes

Upon successful conclusion of this subject, a student will be able to:

1	Create optimal methods for planning, assessing risks and managing analytics projects
2	Justify the rationale behind data analytics projects in business
3	Compare the effectiveness of agile and traditional implementation methods for big data projects
4	Investigate leadership skills in project management



DATA4600 Roadmap

Week 1

Nature of
Analytics Projects

Week 2

Application of Project
Management
methodologies to
Analytics

Week 3

Planning an
Analytics Project

Week 4

Discovery phase of
Analytics Projects &
accessing data

Week 5

Managing time of
analytics projects

Week 6

Managing risk of
Analytics projects

Week 7

Project version
control using Git

Week 8

Scaling an
Analytics Project &
managing changes

Week 9

Leading
Analytics Projects

Week 10

Analytics project
handover

Week 11

Role of AI in
analytics projects

Week 12

Hands-on
simulation



Today's agenda

- Steps in developing a Project Time Plan
- Manage Project Time – Activity Estimation
- Gantt chart
- Network Diagram (PERT chart)
- Kanban Board
- Mind maps
- Precedence Diagramming Method
- Critical Path Analysis
- Activity sequencing
- Lead & Lag Dependencies
- Time contingencies
- Project Libre (open source version of Microsoft Project)



Example of Project Management with data

Sarah's Project

“Sarah’s Smart Path to a Personal Loan (2025)” continued

Now that Sarah is actively managing her personal loan project with Trello and seen the output below, she wonders whether a Kanban Board may be of greater assistance with visualizing the stages of her project?



To Do	In Progress	Waiting for Response	Completed

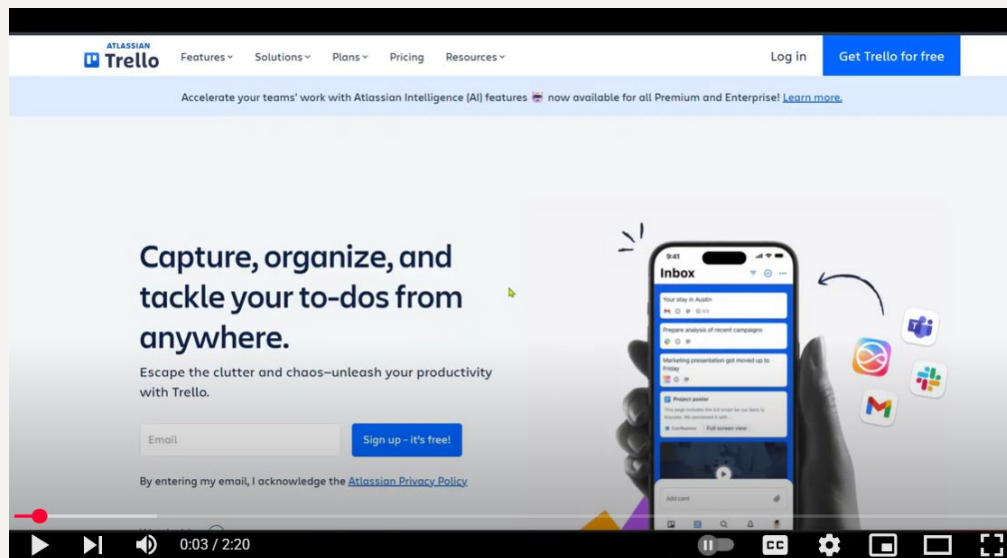
Research lenders	Submit form	Await bank response	Credit score
Gather documents			Docs verified
Choose lender			Loan approved

Example of Project Management with data

Sarah's Kanban Board

- Watch the following video to assist Sarah with creating a Kanban Board in Trello

<https://www.youtube.com/watch?v=RDyotH7rVd4>





Example of Project Management with data

*Sarah's
Project*

Sarah's Personal Loan Kanban Board

Now that Sarah understands how to do a Kanban Board in Trello, here are her lists to include on the Board:

To Do

These are tasks Sarah identified before starting her loan application:

-  **Research Loan Options**
"I want the best interest rate and flexible terms. Let's compare three banks."
-  **Check Credit Score**
"My score will determine how much I can borrow. Time to find out!"
-  **Gather Financial Documents**
"I'll need ID, tax returns, payslips, and proof of address."
-  **Choose Preferred Lender**
"Once I have all the info, I'll decide who to go with."





Example of Project Management with data

*Sarah's
Project*

Sarah's Personal Loan Kanban Board

Now that Sarah understands how to do a Kanban Board in Trello, here are her lists to include on the Board:

In Progress

Tasks Sarah is actively working on:

-  **Filling Out Loan Application**
"The form is long, but I'm halfway through. Uploading docs as I go."
-  **Verifying Details with Lender**
"Spoke with a loan officer—they're checking my employment history."





Example of Project Management with data

*Sarah's
Project*

Sarah's Personal Loan Kanban Board

Now that Sarah understands how to do a Kanban Board in Trello, here are her lists to include on the Board:

Waiting for Response

Sarah is waiting on others before she can act:

-  **Application Under Review**
"They said I'll hear back in 3–5 business days. Fingers crossed!"
-  **Income Confirmation Email**
"The bank emailed me for one more pay stub—I just sent it."





Example of Project Management with data

*Sarah's
Project*

Sarah's Personal Loan Kanban Board

Now that Sarah understands how to do a Kanban Board in Trello, here are her lists to include on the Board:

Tasks Sarah has finished—she's getting closer to her goal:

- ☒ **Checked Credit Score (720)**
"Looks like I'm eligible for a good rate!"
- ☒ **Uploaded Documents**
"All files submitted—on to the next step."
- ☒ **Application Approved**
"YES! Got the approval email today!"
- ☒ **Signed Loan Agreement**
"Read the terms carefully, then signed digitally."
- ☒ **Funds Disbursed**
"\$15,000 in my account. Time to renovate my office!"





Example of Project Management with data

Sarah's Personal Loan Kanban Board

*Sarah's
Project*

Discuss the benefits of Sarah using a Kanban Board for her personal loan project



Steps in developing a Project Time Plan

PMBOK nominates the following steps for managing time in a Business Analytics project or indeed any project



Source: PMBOK



Project Management time management

A visual management tool serves as the visual backbone of project management, allowing project managers, teams, and stakeholders to:

- **Visualize Timelines:** Understand project timelines, including start and end dates, task durations, and milestones.
- **Manage Tasks:** Organize and manage tasks efficiently, assigning responsibilities and dependencies.
- **Track Progress:** Monitor project progress in real-time, identifying delays or bottlenecks.
- **Communicate Effectively:** Facilitate clear communication among team members and stakeholders.
- **Adapt to Changes:** Respond to changing project requirements and adjust timelines or tasks accordingly

<https://www.cq-business-management-software.com/blog/gantt-charts-vs-other-project-visualization-tools-a-comparative-analysis/>

Manage Project Time – Activity Estimation

The following steps assist with allocating time to each of the steps in a project:

- Determine resource requirements/capability
- Apply history
- Assess current market conditions
- Apply subject matter expert judgment
- Consult stakeholders re constraints
- Document assumptions
- Allocate contingencies

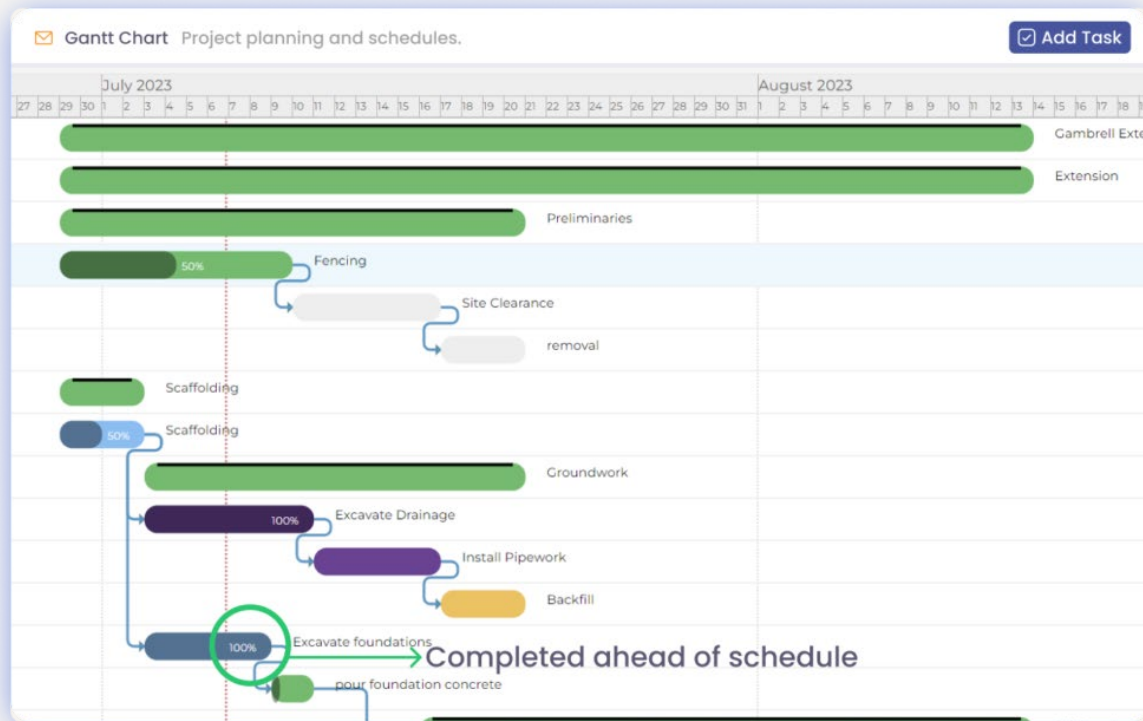


What is a Gantt chart?

- A Gantt chart is a bar chart that shows a project's timeline, including tasks, start and end dates, milestones, dependencies, and assignees.
- The left side of the chart lists tasks, while the right side shows a timeline with horizontal bars, called Gantt bars, that represent each task's duration.
- The Gantt chart is used to monitor progress, keep tasks on track, find the critical path, which is the longest path from project start to completion and provides a visual overview of what work needs to get done, who's doing it, and when.
- Henry Gantt, a social scientist and management consultant who developed the chart in the 1910s.

GANTT charts

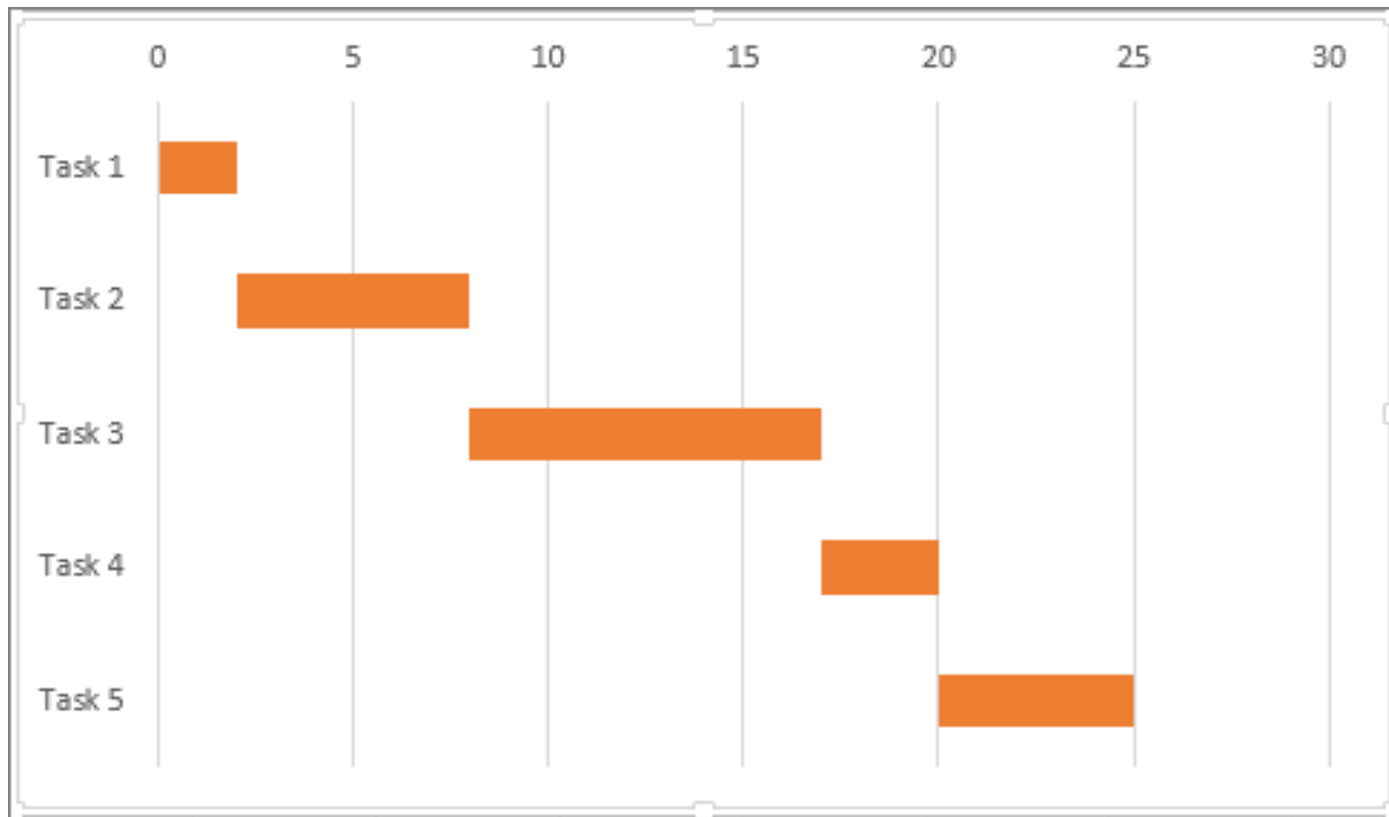
- **Visual Representation:** Gantt charts provide a visual representation of project tasks and timelines
- **Task dependencies:** showcase which tasks must be completed before others can begin.



<https://www.cq-business-management-software.com/blog/gantt-charts-vs-other-project-visualization-tools-a-comparative-analysis/>

Manage Project Time - GANTT chart

Once activities have been estimated, a timeline can be produced with a Gantt chart

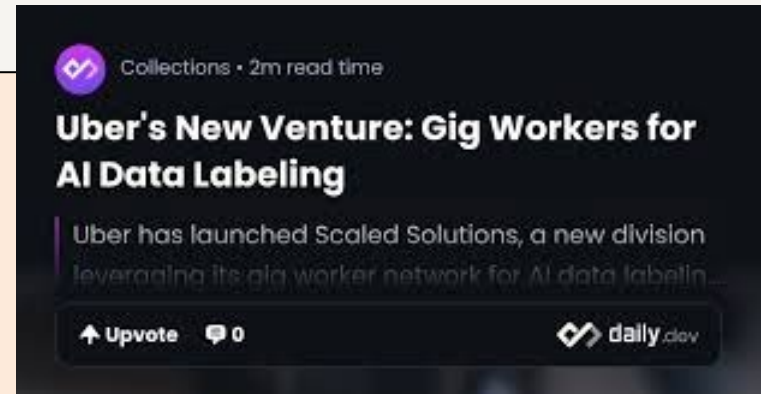


Activity: Uber Data Labelling Project

Form Groups

As the Project Manager for Uber Data Labelling:

-Discuss all the tasks to be included in the Gantt chart.



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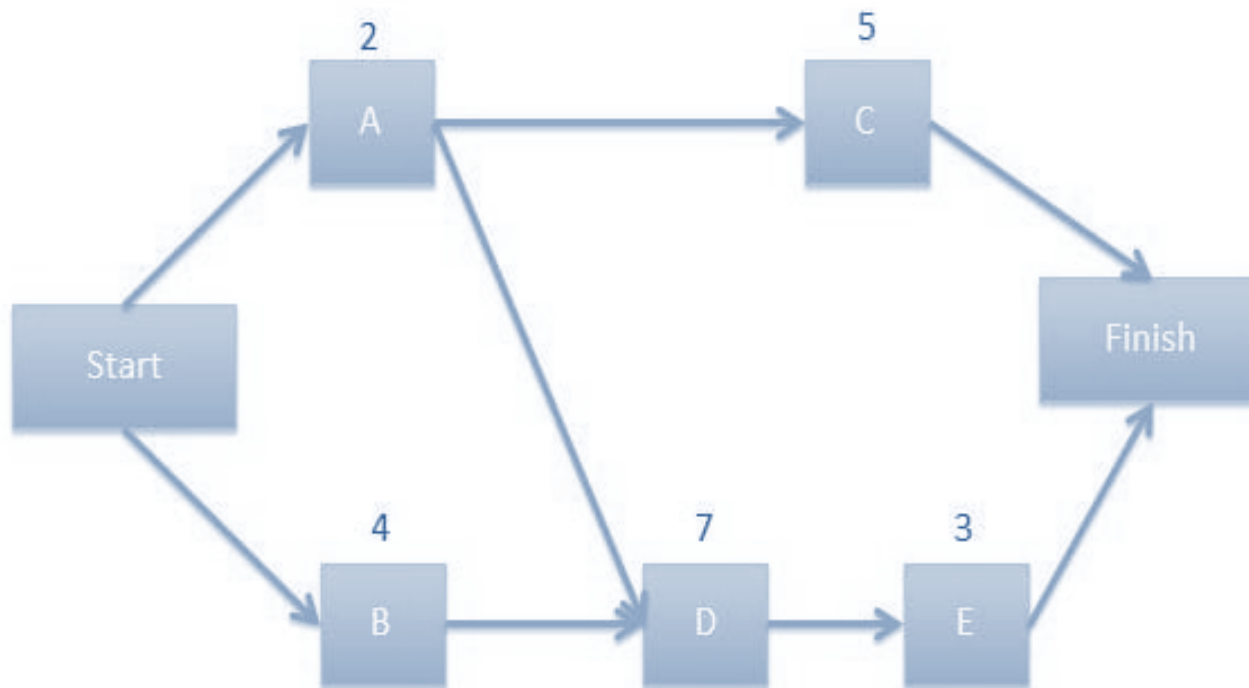


Alternatives to GANTT charts

- **PERT Charts:** Program Evaluation and Review Technique (PERT) charts focus on depicting task dependencies in a network diagram. They are particularly useful for complex projects with numerous dependencies.
- **Kanban Boards:** Kanban boards are popular for their simplicity. They visualize tasks in columns (e.g., To-Do, In Progress, Done) and are excellent for managing workflow.
- **Mind Maps:** Mind maps help with brainstorming and organizing ideas, making them ideal for initial project planning and ideation phases.
- **Precedence Diagramming Method:** helps identify and visualize the relationships between tasks in a project. It uses a network diagram to show how tasks are dependent on each other, and which tasks need to be completed before others can start.

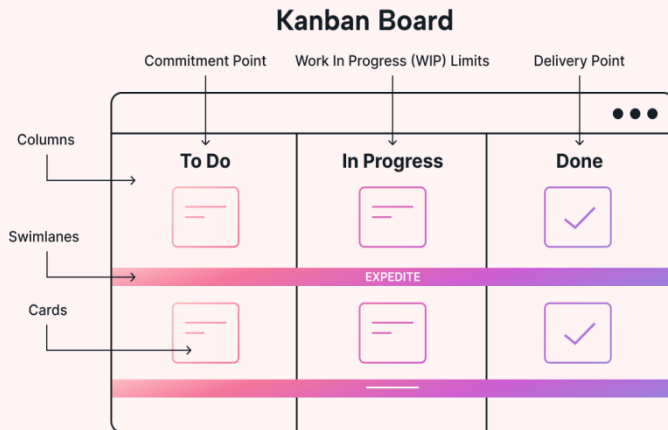
<https://www.cq-business-management-software.com/blog/gantt-charts-vs-other-project-visualization-tools-a-comparative-analysis/>

Network Diagram (PERT chart)



Kanban Board

- A kanban board is an agile project management tool designed to help visualize work, limit work-in-progress, and maximize efficiency (or flow).



Requested	To Do	Coding	Code Review	Testing	Deployment	Done
Messaging on Android Application	App Z Integration	Real-time analytics	Drag-and-drop editor for reports	Switching to SQLite	Autosave feature	Responsive menu
Tagging function						Homepage animations
						Bugged graph visualization

<https://www.usemotion.com/blog/kanban-board-example>

Use of Kanban Boards to assist with Analytics Projects



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- Kanban means “Visual signal” in Japanese
- Kanban is a derivative of lean management
- Work is represented physically/digitally on a **Kanban board** so the status of project tasks is transparent
- Each column is a particular activity or status of such, e.g. “To do”, “Ongoing”, “Complete”
- Column limits may be put in place, e.g.
 - Max of four “ongoing” tasks,
 - Max of three “To do” tasks
- All Kanban projects have a delivery point, e.g. product reaches customer

Activity : Applying Kanban to Analytics projects

Consider development of the pharmacy app for patients, which allows them to order medicines online at the best prices.

Form groups and consider the following questions:

Q1. Summarise the idea of Kanban

Q2. Create a Kanban board for the pharmacy app development project.



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Q3. Why might the Kanban framework fail in the pharmacy app case?



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When Kanban Fails

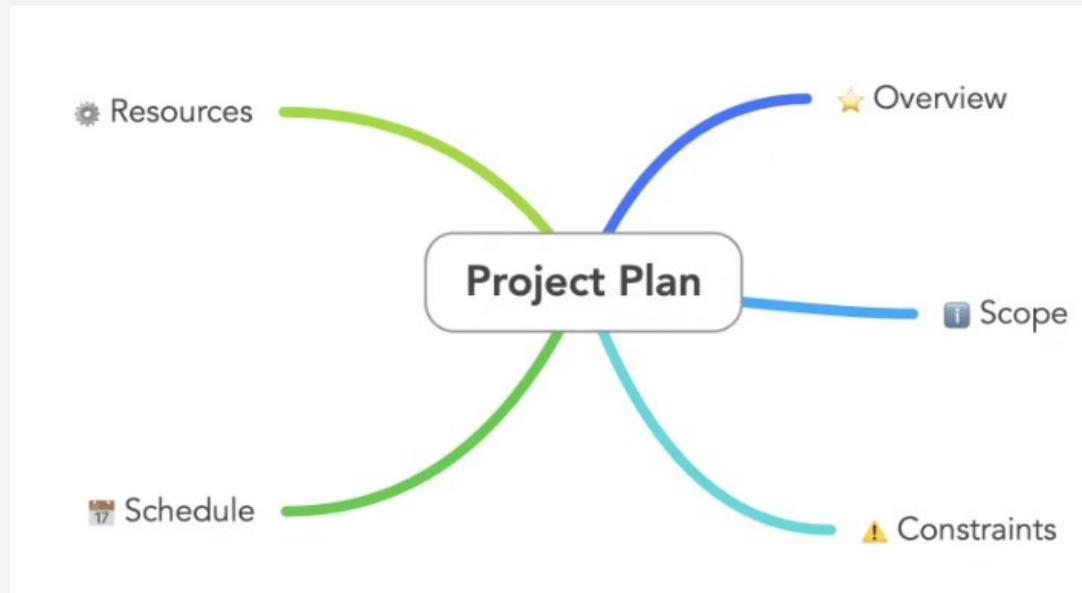
- The board is not kept up to date
- Posting what you think the team wants to see on the board rather than reality (imagined processes)
- Overly complex boards
- Overlooking/ignoring column limits
- Not realising that Kanban does not have specific rules to make a process work, rather it is more about workflow

<https://kanbantool.com/kanban-library/kanban-results/when-kanban-fails>



Mind maps

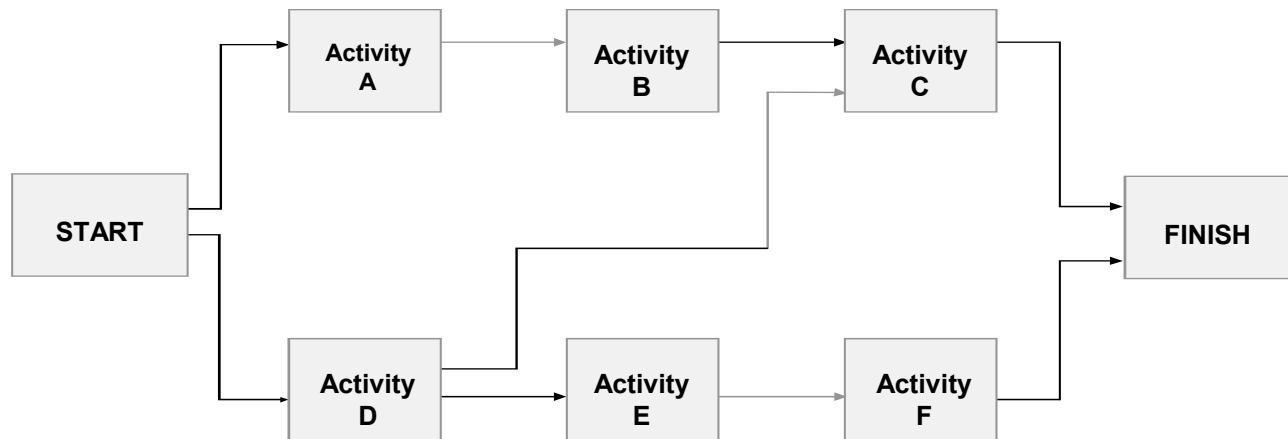
- A mind map visualizes ideas and concepts, providing a systematic way to capture and organize thoughts.
- It helps users understand concepts by breaking them down into their component parts.
- It helps teams generate ideas and better understand and organize existing information.



<https://www.mindmaps.com/project-planning-with-mind-maps/>

Precedence Diagramming Method

The precedence diagramming method (PDM) is a graphical tool that uses boxes, or nodes, to represent activities and arrows to show dependencies. It helps prevent scheduling conflicts, minimizes downtime, and enhances team productivity





GANTT charts versus Alternatives

Visualization Style

- **Gantt Charts:** Gantt charts provide a linear, time-based view of tasks and dependencies. They offer a clear, structured overview of a project.
- **Alternatives:** Kanban boards offer a more fluid, column-based view, while PERT charts and mind maps emphasize visualizing task relationships in non-linear ways.

Use Cases

- **Gantt Charts:** Ideal for projects with well-defined timelines and dependencies, such as construction projects and product launches.
- **Alternatives:** Kanban boards are great for ongoing, iterative projects, PERT charts for complex projects, and mind maps for brainstorming and ideation.

Task and Dependency Management

- **Gantt Charts:** Excel in depicting task dependencies, making them suitable for projects with intricate task relationships.
- **Alternatives:** Kanban boards are more task-centric, PERT charts excel in visualizing complex dependencies, and mind maps focus on idea organization.



When to use GANTT chart versus alternatives

Choosing the Right Tool

- Use **Gantt Charts** for well-structured projects with clear timelines and dependencies.
- Choose **Kanban Boards** for ongoing, iterative projects that benefit from visualizing workflow.
- Opt for **PERT Charts** when managing highly complex projects with intricate task dependencies.
- Embrace **Mind Maps** during brainstorming and ideation phases, or for organizing ideas before formal project planning.

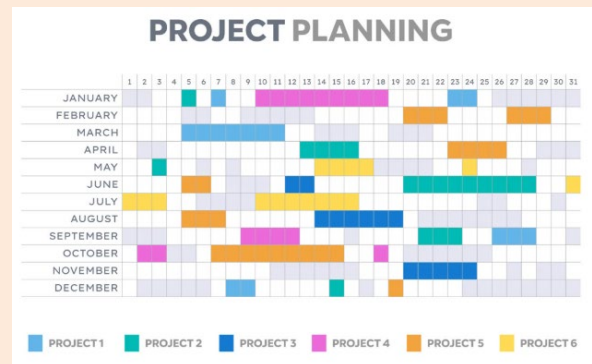
<https://www.cq-business-management-software.com/blog/gantt-charts-vs-other-project-visualization-tools-a-comparative-analysis/>

Activity: Uber Data Labelling Project cont)

Form Groups

For the Uber Data Labelling project, nominate which project time management tool you would find more useful

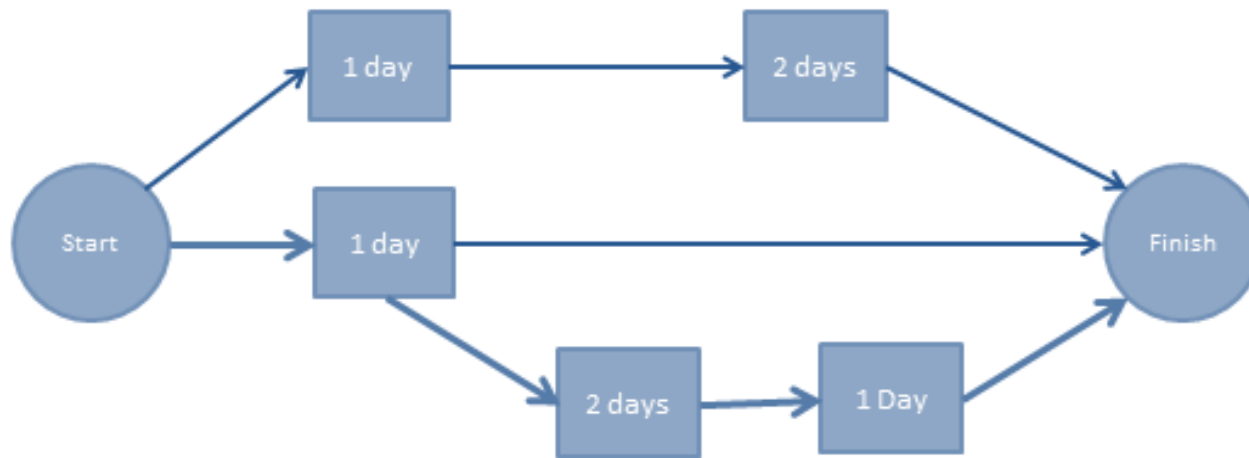
Discuss your view with the rest of the class



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Manage Project Time- Critical Path Analysis

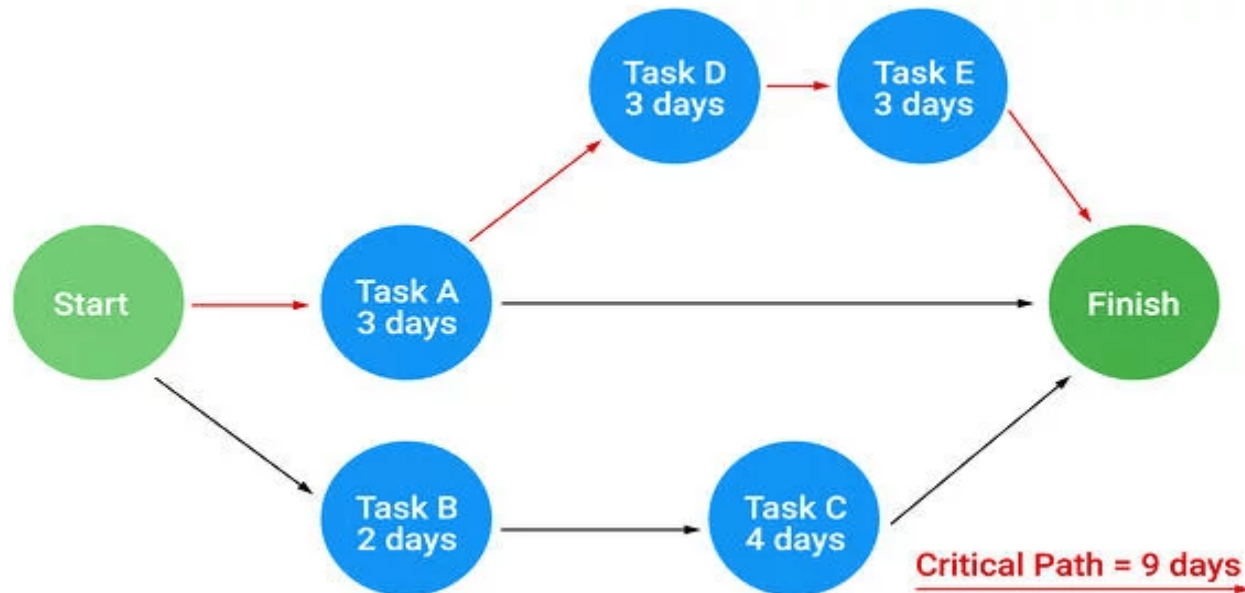
- The **critical path** method is a schedule network analysis technique that calculates the earliest project completion date
- The longest time to complete the task



Critical Path = 4 Days

Manage Project Time- Critical Path Analysis

The critical path is the longest time to complete the task



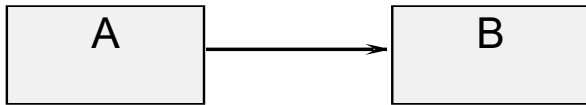
Manage Project Time- Activity Sequencing

Four types of dependencies or precedence relationships:

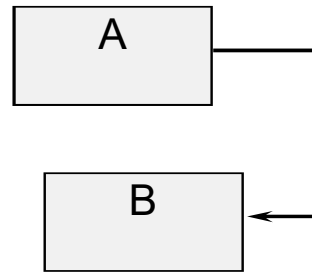
- **Finish-to-Start (FS).** This relationship means Task A must complete before Task B can begin. This is the most common relationship.
- **Finish-to-Finish (FF).** This relationship means Task A must complete before Task B does.
- **Start-to-Start (SS).** This relationship means Task A must start before Task B can start.
- **Start-to-Finish (SF).** This relationship is unusual and is rarely used. It requires that Task A start so that Task B may finish.

Manage Project Time – Dependencies

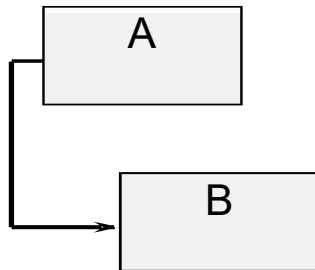
1. Finish to Start



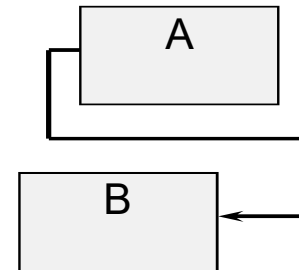
3. Finish to Finish



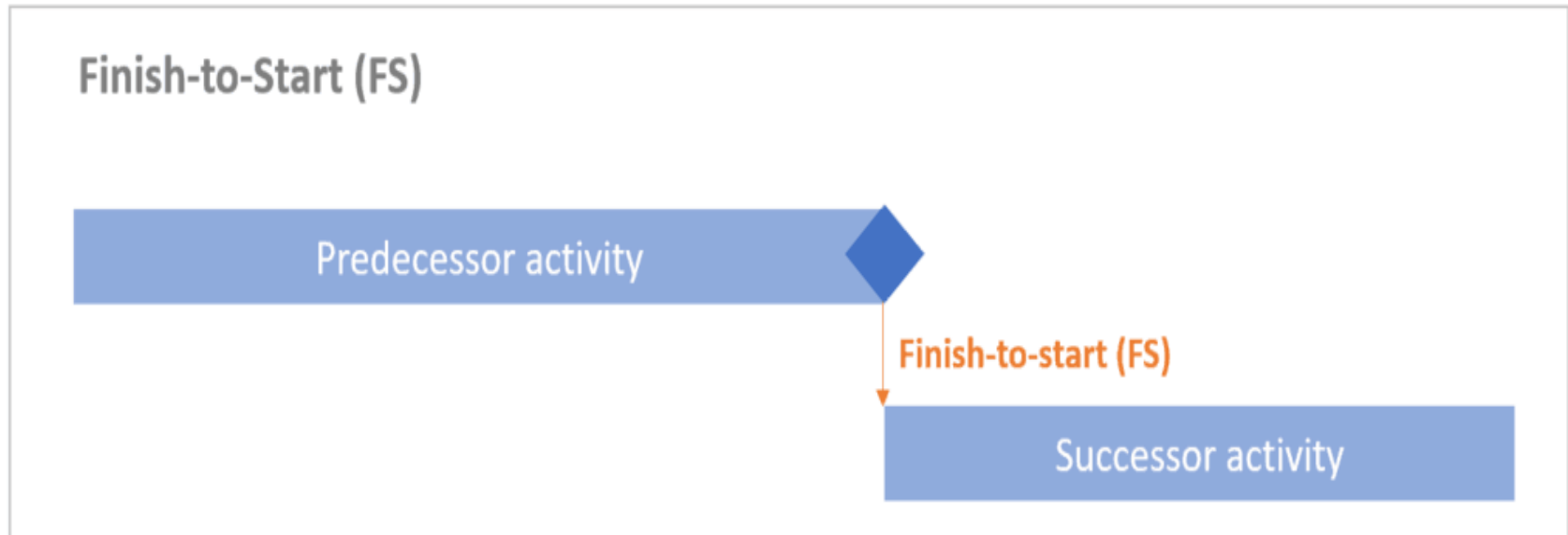
2. Start to Start



4. Start to Finish



Manage Project Time- Activity Sequencing

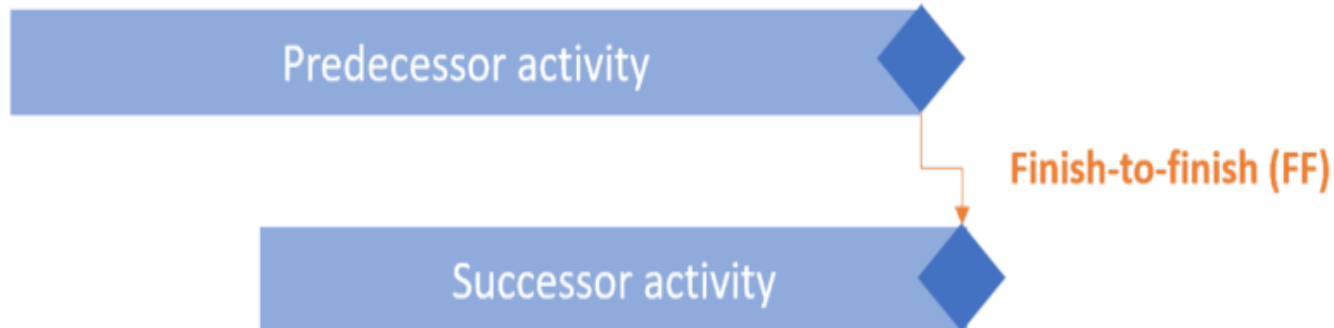


Project-Management.info

- When the cooking is finished, then dinner commences
- When the data is cleansed, it can be input to the data model

Manage Project Time- Activity Sequencing

Finish-to-Finish (FF)



Project-Management.info

- If you have rented a lab for a science experiment, the science experiment must be complete when the lab rental ends.
- The data model must be debugged by the time the project is finished and passed to the BAU team

Manage Project Time- Activity Sequencing

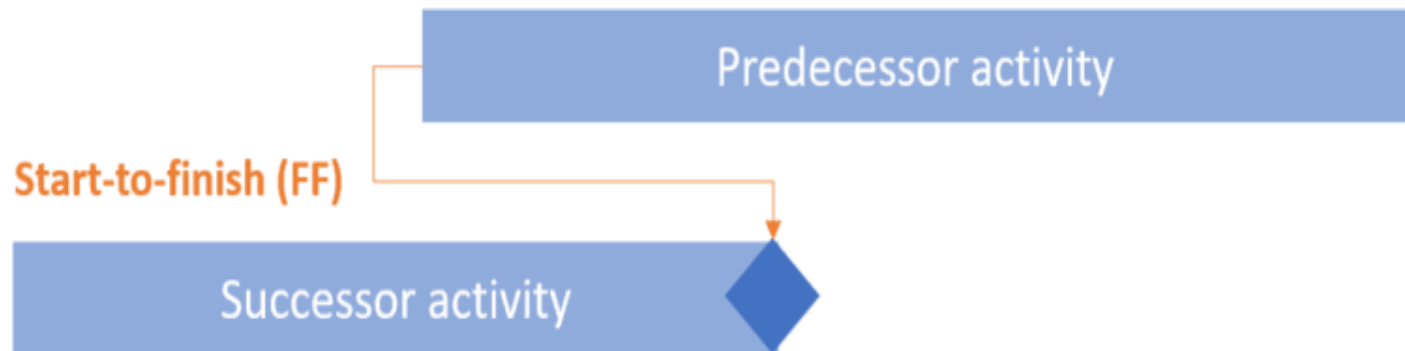


Project-Management.info

- When the research paper commences, then editing can commence
- When the data is accessed, the data can commence being cleaned

Manage Project Time- Activity Sequencing

Start-to-Finish (SF)



Project-Management.info

- You can only start to deactivate the old IT system (predecessor) once the new IT system has been enabled and users can work with it (successor).



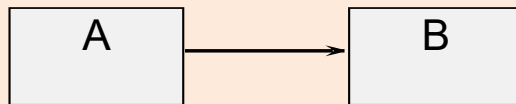
Activity: Uber Data Labelling Project cont)

Form Groups

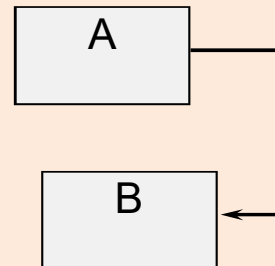
For the Uber Data Labelling project, nominate activities which belong to each of the time sequencing activities.

Discuss your findings with the rest of the class

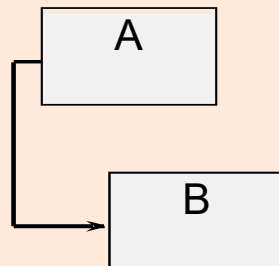
1. Finish to Start



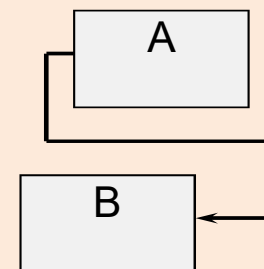
3. Finish to Finish



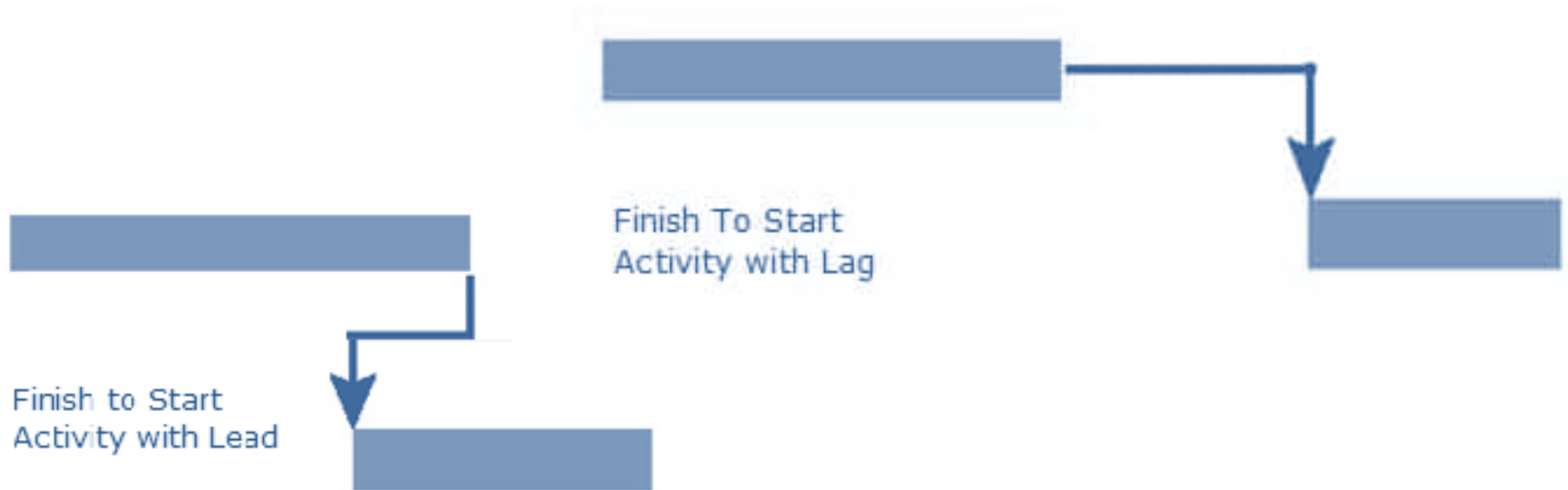
2. Start to Start



4. Start to Finish



Manage Project Time – Lead & Lag Dependencies



Lead

The amount of time a successor task can start before its predecessor is completed. Leads allow tasks to overlap, which can shorten project duration.

Lag

The amount of time that must pass before a successor task can start after its predecessor is completed.

Lags introduce delays between tasks to ensure dependencies are managed effectively.

Manage Project Time – Contingency

- Effort contingency – additional work/cost in plan for:
 - overlooked tasks
 - inaccurate estimates
- Elapsed time or schedule contingency
 - allow for lower productivity and “learning curve”

Manage Project Time – Utilise Project Libre

Project Libre provides an open source alternative to Microsoft Project

ProjectLibre™



Project Libre is a free, open-source project management software, functioning as an alternative to Microsoft Project.

Project Libre Desktop

- It's available for anyone to download and use without cost.
- Includes essential project management tools like Gantt charts, network diagrams, resource management, and work breakdown structures (WBS).

Project Libre Cloud

- Subscription-based
- Offers a cloud-based platform with real-time collaboration features for teams.
- Utilises artificial intelligence for tasks like generating project plans from natural language prompts.



Activity: Utilisation of Project Libre

Your facilitator will demonstrate how to use Project Libre

Apply Project Libre to the activity sequencing you nominated in your prior activity today.

Download Project Libre Desktop

<https://www.projectlibre.com/>

The Project Libre logo, featuring the word "Project" in red and "Libre" in black, with a small trademark symbol.

Step 1: Create a New Project

- Open Project Libre.
- Click on **File** → **New**.
- Enter:
 - Project Name
 - Start Date
 - Manager Name
 - Calendar settings (default or custom workdays)
- Click **OK**.

Step 2: Add Tasks

- Go to the **Tasks** tab.
- Enter task names in the **Task Name** column.
- Add **durations** (e.g., 3d, 1w).
- To create a hierarchy use **indent/outdent** for sub-tasks.
- To link tasks (dependencies) use the **Predecessors** column



ProjectLibreTM

Step 3: Create and Assign Resources

- Switch to **Resources** tab.
- Enter:
 - **Resource Name**
 - **Type** (Work or Material)
 - **Standard Rate** (for cost tracking)
 - **Max Units** (percentage availability)
- Return to **Gantt Chart** view.
- Open **Task Information** (double-click a task).
- Go to **Resources** tab and assign appropriate resources.

Step 4: Set a Baseline

- Ensure your schedule is complete (tasks, durations, dependencies, resources).
- Click on **Project → Set Baseline**.
- Confirm to save the current plan as your baseline.
- This allows comparison between planned and actual progress

Step 5: Track Progress

- As the project progresses, update:
 - % Complete
 - Actual Start/Finish Dates
 - Actual Work
- Project Libre will show variance from the baseline



Next week

- Managing risks in Analytics projects



References

CPMP 2020, *COVID19 Global Forecasting (Week 5)*, Kaggle, viewed 08 January 2024
<https://www.kaggle.com/c/covid19-global-forecasting-week-5/discussion/151461>